



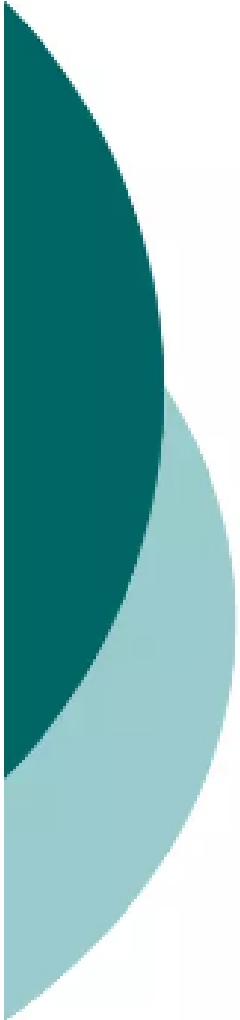
Mathematical Logic

- Definition: Methods of reasoning, provides rules and techniques to determine whether an argument is valid
- Theorem: a statement that can be shown to be true (under certain conditions)
 - Example: If x is an even integer, then $x + 1$ is an odd integer
 - This statement is true under the condition that x is an integer is true



Mathematical Logic

- A **statement**, or a **proposition**, is a declarative sentence that is either true or false, but not both
- Uppercase letters denote propositions
 - Examples:
 - P: 2 is an even number (true)
 - Q: 7 is an even number (false)
 - R: A is a vowel (true)
 - The following are not propositions:
 - P: My cat is beautiful
 - Q: My house is big



Propositions

- A statement that has a truth value
- Which of the following are propositions?
 - The Washington State flag is red
 - It snowed in Whistler, BC on January 4, 2008.
 - Hillary Clinton won the democratic caucus in Iowa
 - Space aliens landed in Roswell, New Mexico
 - Ron Paul would be a great president
 - Turn your homework in on Wednesday
 - Why are we taking this class?
 - If n is an integer greater than two, then the equation $a^n + b^n = c^n$ has no solutions in non-zero integers a , b , and c .
 - Every even integer greater than two can be written as the sum of two primes
 - This statement is false
- Propositional variables: $p, q, r, s, \dots$
- Truth values: **T** for true, **F** for false



Mathematical Logic

- Truth value
 - One of the values "*truth*" (T) or "*falsity*" (F) assigned to a statement
- Negation
 - The **negation** of P , written $\neg P$, is the statement obtained by negating statement P
 - *Example:*
 - P : A is a consonant
 - $\neg P$: it is the case that A is not a consonant
- Truth Table

P	$\neg P$
T	F
F	T

Mathematical Logic

Conjunction

- Let P and Q be statements. The **conjunction** of P and Q , written $P \wedge Q$, is the statement formed by joining statements P and Q using the word “and”
- The statement $P \wedge Q$ is true if both p and q are true; otherwise $P \wedge Q$ is false
- Truth Table for Conjunction:

<u>P</u>	<u>Q</u>	<u>P $\wedge$ Q</u>
F	F	F
F	T	F
T	F	F
T	T	T

Mathematical Logic

- Disjunction

- Let P and Q be statements. The **disjunction** of P and Q , written $P \vee Q$, is the statement formed by joining statements P and Q using the word "or"
- The statement $P \vee Q$ is true if at least one of the statements P and Q is true; otherwise $P \vee Q$ is false
- The symbol $\vee$ is read "or"
- Truth Table for Disjunction:

P	Q	$P \vee Q$
F	F	F
F	T	T
T	F	T
T	T	T

Exclusive disjunction

- “Either P or Q” (but not both), in symbols $P \underline{\vee} Q$

P	Q	$P \underline{\vee} Q$
T	T	F
T	F	T
F	T	T
F	F	F

- $P \underline{\vee} Q$ is true only when P is true and Q is false, or P is false and Q is true.
 - Example: p = "John is programmer, q = "Mary is a lawyer"
 - $p \underline{\vee} q$ = "Either John is a programmer or Mary is a lawyer"



Mathematical Logic

○ Implication

- Let P and Q be statements. The statement "if P then Q " is called an **implication or condition**.
- The implication "if P then Q " is written $P \rightarrow Q$
- P is called the hypothesis, Q is called the conclusion

○ Truth Table for Implication:

<u>P</u>	<u>Q</u>	<u>$P \rightarrow Q$</u>
F	F	T
F	T	F
T	F	T
T	T	T



Mathematical Logic

- Implication

- Let P : Today is Sunday and Q : I will wash the car.
- $P \rightarrow Q$:
If today is Sunday, then I will wash the car
- The **converse** of this implication is written $Q \rightarrow P$
If I wash the car, then today is Sunday
- The **inverse** of this implication is $\neg P \rightarrow \neg Q$
If today is not Sunday, then I will not wash the car
- The **contrapositive** of this implication is $\neg Q \rightarrow \neg P$
If I do not wash the car, then today is not Sunday

Mathematical Logic

Biimplication

- Let P and Q be statements. The statement " P if and only if Q " is called the **biimplication or biconditional** of P and Q
- The biconditional " P if and only if Q " is written $P \leftrightarrow Q$
- " P if and only if Q "
- Truth Table for the Biconditional:

P	Q	$P \leftrightarrow Q$
F	F	T
F	T	F
T	F	F
T	T	T

Propositional Equivalences

The compound proposition p and q is said to be Logical equivalences if $p \leftrightarrow q$ is a Tautology . Logical equivalence is denoted by $\equiv$ and $\iff$

TABLE 6 Logical Equivalences.	
<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

TABLE 7 Logical Equivalences Involving Conditional Statements.
$p \rightarrow q \equiv \neg p \vee q$ $p \rightarrow q \equiv \neg q \rightarrow \neg p$ $p \vee q \equiv \neg p \rightarrow q$ $p \wedge q \equiv \neg(p \rightarrow \neg q)$ $\neg(p \rightarrow q) \equiv p \wedge \neg q$ $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$ $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$ $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$ $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

TABLE 8 Logical Equivalences Involving Biconditional Statements.
$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$ $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$ $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

Equivalence	Name of Identity
$p \wedge T \equiv p$ $p \vee F \equiv p$	Identity Laws
$p \wedge F \equiv F$ $p \vee T \equiv T$	Domination Laws
$p \wedge p \equiv p$ $p \vee p \equiv p$	Idempotent Laws
$\neg(\neg p) \equiv p$	Double Negation Law
$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$	Commutative Laws
$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$	Associative Laws
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	Distributive Laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's Laws
$p \wedge (p \vee q) \equiv p$ $p \vee (p \wedge q) \equiv p$	Absorption Laws
$p \wedge \neg p \equiv F$ $p \vee \neg p \equiv T$	Negation Laws

$$p \vee (p \wedge q) = (p \vee p) \wedge (p \vee q)$$

[using Distributive law]

$$= p \wedge (p \vee q)$$

[$\because p \vee p = p$]

$$= (p \wedge p) \vee (p \wedge q)$$

[using Distributive law]

$$= p \vee (p \wedge q)$$

[$\because p \wedge p = p$]

$$(a) \quad p \wedge (1 \vee q) = p$$

[Taking p as common]

$$(a) \quad p \vee 1 = 1$$

[$\because 1 \vee q = 1$]

$$(a) \quad p \vee p = p$$

[$\because p \wedge 1 = p$]

$\neg(p \vee (\neg p \wedge q))$	$\equiv \neg p \wedge \neg(\neg p \wedge q)$	<i>Using De Morgan's law</i>
	$\equiv \neg p \wedge (\neg(\neg p) \vee \neg q)$	<i>Using De Morgan's law</i>
	$\equiv \neg p \wedge (p \vee \neg q)$	<i>Using Double negation law</i>
	$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	<i>Using Distributive law</i>
	$\equiv F \vee (\neg p \wedge \neg q)$	<i>Using Negation Law</i>
	$\equiv \neg p \wedge \neg q$	<i>Using Identity Law</i>